1. Clinical Question & Core Hypothesis

Clinical Question

Tumor-stroma ratio (TSR) is an image-derived prognostic signal in NMIBC, but why stroma-rich tumors more frequently progress after TURBT is unclear — does the stromal compartment actively enable residual tumor survival, invasion, or immune evasion that drives progression? 2) Data basis / paths used: only a single, morphology→outcome path was provided (TSR → NMIBC prognosis), so I built the hypothesis directly on that mechanistic link: TSR as a readout of a pro-invasive, immune-excluding microenvironment. 3) Supporting evidence (Tier 2 style): spatial morphologic phenomena (tumor budding, CAF-rich ECM, and lymphocyte exclusion) conceptually support that high stromal fraction can create physical/biochemical niches favoring residual cell survival and local invasion after resection. 4) Endpoint choice rationale: Progression-free interval (PFI) is the clinically decisive outcome in NMIBC (transition to MIBC drives morbidity and treatment change), so it is the most relevant and measurable endpoint for a TSR-based prognostic mechanism. 5) Deliberate exclusions: No molecular anchor genes were included because the provided input contained only morphologic/spatial nodes (introducing specific genes would invent data not supplied). Conventional clinicopathologic covariates (grade, stage Ta/T1, multiplicity, prior recurrences, adjuvant intravesical therapy) are placed as secondary modifiers to be adjusted for in analyses?

Working Hypothesis

Primary hypothesis: In NMIBC patients treated with TURBT, a high tumor stroma ratio (TSR) in diagnostic tumor sections is independently associated with a shorter progression-free interval (PFI) owing to an activated stromal microenvironment that (i) promotes residual tumor cell survival and local invasion through CAF-driven ECM remodeling, and (ii) facilitates immune exclusion, reducing anti-tumor immune responses. Consequently, TSR provides incremental prognostic value beyond conventional clinicopathologic factors, improving risk stratification for progression when incorporated into a multivariable model focused on PFI.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Primary endpoint	PFI

2. Why This Gene — Not Others

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	grade	Supporting modifier from knowledge graph
Secondary	stage (Ta/T1)	Supporting modifier from knowledge graph

Secondary	tumor multiplicity	Supporting modifier from knowledge graph
Secondary	prior recurrences	Supporting modifier from knowledge graph
Secondary	adjuvant intravesical therapy	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- grade: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- stage (Ta/T1): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.